

Symmetric Rigid Body Maneuvering (SRBM)

Top-Level Architectural Specification

Autonomous Air Combat Control System
AI-Native Tactical and Rigid-Body Architecture

Version 1.0

June 4, 2026

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This document defines the complete SRBM architecture, including doctrine modeling, latent tactical intent, rigid-body mapping, envelope enforcement, allocation, and tactical cost and reward frameworks.

Abstract

The Symmetric Rigid Body Maneuvering (SRBM) architecture defines a top-level, airframe-agnostic control and tactical decision framework for AI-native air combat systems. The architecture requires that the physical airframe present itself to the tactical layer as an idealized, symmetric, 6-DoF rigid-body entity operating within an abstract geometric space. Layers 1 and 2 form a high-rate virtualization substrate that guarantees this abstraction, ensuring that the tactical layer never interacts with aerodynamic modes, stability derivatives, or lift-vector assumptions.

SRBM establishes a complete architectural stack spanning doctrine-level tactical intent generation, latent-space blending, rigid-body command synthesis, envelope enforcement, and actuator-level moment allocation. A strict multi-rate hierarchy is imposed, requiring the control substrate to operate at a frequency

$$f_{\text{ctrl}} \geq K \cdot f_{\text{AI}}, \quad K \geq 20,$$

so that all aerodynamic transients, actuator nonlinearities, and moment-allocation perturbations are fully resolved within a single tactical epoch.

As a top-level specification, SRBM formalizes the interfaces, mathematical structures, and behavioral contracts governing the three SRBM layers: tactical decision making, envelope protection, and high-authority actuation. Detailed aerodynamic models, state-estimation filters, adversarial uncertainty models, and actuator-level control laws are intentionally out of scope and are supplied by lower-level subsystem specifications beneath SRBM.

The observation tensor includes a filtered execution-gap signal O_{gap}^t , which serves as the formal abstraction boundary between physical reality and tactical cognition. By exposing only this bounded, low-frequency performance-clamping signal, SRBM eliminates the traditional Sim-to-Real gap and ensures that the AI adapts to deterministic behavior rather than raw turbulence or actuator noise.

This document defines the observation tensors, latent intent representations, and tactical cost and reward functions used for deterministic execution and reinforcement-learning-based training. The SRBM framework provides a coherent, modular foundation for predictable, interpretable, and tactically effective maneuvering behavior in high-dynamics air combat environments.

Nomenclature

Symbol	Units	Meaning
q_{WB}	unit quaternion	World-to-body attitude quaternion
R_{WB}	SO(3)	Rotation matrix derived from q_{WB}
$\mathbf{v}^w$	—	Tactical vector in world frame
$\mathbf{v}^b$	—	Tactical vector in body frame ($\mathbf{v}^b = R_{WB}\mathbf{v}^w$)
p, q, r	[-1,1]	Roll, pitch, yaw rate commands (signed)
T, B	[0,1]	Thrust and brake commands (nonnegative)
$\mathbf{u}_{RB}$	—	Proposed rigid-body command vector (p, q, r, T, B)
$\mathbf{u}_{RB, safe}$	—	Envelope-safe rigid-body command (Layer 2 output)
$\mathbf{u}_i$	[-1,1]	Tactical intent vector for anchor $i \in \{HO, HD, DO, DD\}$
$\mathbf{u}_{tac}^w$	—	Blended world-frame tactical intent
$\mathbf{u}_{tac}^b$	—	Body-frame tactical intent after rotation
w_i	[0,1]	Anchor blending weight, $\sum_i w_i = 1$
α	[0,1]	Coordination axis (Hive $\leftrightarrow$ Individual)
γ	[0,1]	Combat posture axis (Defensive $\leftrightarrow$ Offensive)
z_i	[-1,1]	Latent tactical vector for behavior i
z_{tac}	[-1,1]	Blended latent tactical vector
$p_{cmd}, q_{cmd}, r_{cmd}, T_{cmd}, B_{cmd}$		Layer 3 commanded rigid-body vector
$\mathbf{V}^b$	m/s	Body-frame velocity (V_x, V_y, V_z)
AoA	deg/rad	Angle of attack
β	deg/rad	Sideslip angle
q_d	Pa	Dynamic pressure
n	g	Load factor
$\dot{n}$	g/s	Load factor rate
E_s	J/kg	Specific energy
E	[0,1]	Normalized energy advantage
H_{phys}	[0,1]	Physical health
H_{sens}	[0,1]	Sensor health
H_{eff}	[0,1]	Effective health (combined)
c_β, c_E, c_S, c_H	[0,1]	Posture weights (sum to 1)
γ_{raw}	[0,1]	Raw posture estimate (pre-logistic)
τ	>0	Temporal filter constant
$\sigma(u)$	[-1,1]	Smooth saturation function (e.g., $\tanh(ku)$)
$p_{raw}, q_{raw}, r_{raw}$	[-1,1]	Raw angular commands
T_{raw}, B_{raw}	[0,1]	Raw energy commands
G_p, G_q, G_r	[0,1]	Authority fractions for angular channels

Symbol	Units	Meaning
G_T, G_B	[0,1]	Authority fractions for energy channels
x	—	State vector to envelope networks
B_x	—	Control effectiveness matrix
r_{mom}	N·m	Desired moment vector
δ_i	deg/rad	Actuator deflection
$\mathbf{u}_{act}$	—	Actuator command vector
$B^\dagger$	—	Pseudoinverse of B_x
a_t	—	RL action (p, q, r, T, B)
s_t	—	RL state
R_{DO}, R_{DD}	—	DO/ DD reward functions
$\mathbf{r}_{ij}$	m	Relative position
R_{ij}	m	Range
$\dot{R}_{ij}$	m/s	Closure rate
LOS	—	Line-of-sight vector
$\dot{\text{LOS}}$	rad/s	LOS rate

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Part I

Top-Level Architecture

0.1 SRBM Architectural Overview

The Symmetric Rigid-Body Maneuvering (SRBM) architecture defines an AI-native air combat system built from first principles around high-authority actuation, rigid-body physics, and solver-driven control. Unlike legacy fighter aircraft, whose geometry, stability margins, and control laws are shaped by human physiological limits, SRBM assumes a machine-only operator and removes all human-era constraints.

A core architectural requirement is that the physical airframe present itself to the tactical layer as an idealized, symmetric, 6-DoF rigid-body entity operating within an abstract geometric space. Layers 1 and 2 form a high-rate virtualization substrate that guarantees this abstraction by absorbing aerodynamic nonlinearities, actuator limits, and transient perturbations before they reach the tactical layer. As stated in the Top-Level Specification, the architecture is “defined from first principles around autonomous control, high-authority actuation, and rigid-body maneuvering” [?]. This shift enables maneuver classes and engagement geometries not achievable with traditional lift-vector aircraft.

To ensure that the rigid-body abstraction remains valid, SRBM imposes a strict multi-rate hierarchy between the tactical loop and the control substrate. The control substrate must operate at a frequency

$$f_{\text{ctrl}} \geq K \cdot f_{\text{AI}}, \quad K \geq 20,$$

so that aerodynamic transients, control-surface blanking, and moment-allocation perturbations are fully resolved within a single tactical epoch. This guarantees that the tactical layer observes only the filtered, envelope-safe behavior of the virtualization substrate.

0.1.1 Design Claims and Non-Claims

This section defines the explicit claims made by the SRBM architecture and the limits of those claims. Its purpose is to prevent misinterpretation of SRBM as aerodynamically unconstrained or physically universal.

Explicit Design Claims

SRBM makes the following concrete and defensible claims:

1. **Rigid-body-native tactical representation.** Tactical intent is expressed in a frame-agnostic rigid-body space rather than in pilot-centric abstractions such as bank angle, lift-vector turn, or sustained-G maneuvering.
2. **Doctrine–geometry decoupling.** Tactical behavior is generated in a latent space and blended prior to command synthesis, allowing doctrine to remain independent of airframe geometry.

3. **Envelope-authoritative feasibility enforcement.** Layer 2 defines the feasible maneuver manifold and enforces it strictly, ensuring that all commands respect aerodynamic, structural, and actuator constraints.
4. **Execution-gap feedback.** Layer 3 receives real-time information about clamping, saturation, and authority shortfall, enabling tactical reasoning that is aware of feasibility limits. The execution-gap signal serves as the formal abstraction boundary between physical reality and tactical cognition.

Non-Claims

SRBM does not claim:

- that aerodynamic forces are negligible or secondary,
- that orientation is physically irrelevant to control effectiveness,
- that high-angle-of-attack regimes are well-conditioned or easy to control,
- that the plant behaves as a free rigid body in atmosphere,
- that SRBM constitutes a complete tactical cognition model including EW, communications, or ROE.

The architecture assumes nonlinear aerodynamics, state-dependent control effectiveness, actuator limits, aeroelastic and structural constraints, and poorly conditioned dynamics in extreme regimes. These realities are handled by Layer 2 and the state-dependent allocation layer.

Correct Physical Interpretation

SRBM replaces pilot-centric maneuver abstractions with rigid-body-native maneuver abstractions for high-authority autonomous atmospheric vehicles, while still respecting aerodynamic and structural constraints through Layer 2 feasibility enforcement and execution-gap feedback. The tactical layer interacts only with the virtualization substrate’s filtered, envelope-safe representation of the airframe.

Architectural Innovations

The primary innovations introduced by SRBM are:

1. rigid-body tactical intent without lift-vector assumptions,
2. latent-space doctrine blending prior to command synthesis,
3. envelope enforcement as an authoritative safety manifold,
4. execution-gap feedback as a first-class tactical signal,
5. airframe-agnostic tactical reasoning enabled by the virtualization substrate.

0.1.2 Core Architectural Principles

- **Rigid-body maneuvering:** Orientation is arbitrary; the aircraft is treated as a 6-DoF physics object rather than a lift-dependent airframe [?].
- **Symmetric authority:** The airframe is designed with left/right and dorsal/ventral symmetry, eliminating privileged orientations and enabling full-attitude maneuvering.
- **Layered control:** Tactical autonomy, envelope enforcement, and actuator allocation are separated into three authority layers with clean interfaces.
- **AI-native doctrine:** Tactical behavior is expressed in a continuous doctrine space (coordination α and posture γ), blended from four anchor behaviors [?].
- **Solver-driven actuation:** All control surfaces and thrust-vectoring actuators are commanded through a moment-allocation solver, not pilot-centric control laws.

0.1.3 Operational Consequences

The Top-Level Specification emphasizes that removing human constraints enables:

- full-attitude maneuvering, including post-stall and high-AoA regimes,
- rapid attitude transitions unconstrained by pilot G-limits,
- cooperative swarm geometries enabled by doctrine blending,
- rigid-body tactical representation that does not assume a lift-vector turn model [?].

This architecture defines the foundation for all subsequent tactical, envelope, and allocation layers.

0.2 Airframe Top-Level Specification

The SRBM airframe is designed as a high-authority, fully symmetric maneuvering platform optimized for autonomous control. The Top-Level Specification describes it as “a high-authority maneuvering platform optimized for autonomous control” with symmetric geometry and no privileged lift orientation [?].

0.2.1 Design Philosophy

- **Symmetric geometry:** Zero dihedral, zero geometric camber, and symmetric airfoils produce no inherent lift at zero AoA. Lift is commanded, not inherent [?].
- **Rigid-body orientation:** The airframe has no preferred “upright” state; orientation is arbitrary and enforced by control, not aerodynamics.

- **High-authority actuation:** Control surfaces, thrust vectoring, and braking surfaces provide full 3-axis moment authority.
- **CG-aligned thrust:** The propulsion system is aligned through the center of gravity to minimize coupling.

0.2.2 Reference Configuration

The Appendix A reference configuration describes:

- **X-tail quad-rudder array:** Four identical all-moving canted rudders providing yaw, pitch augmentation, and braking authority [?].
- **Thrust vectoring:** A high-bandwidth nozzle with a $\pm 22^\circ$ vectoring cone.
- **High TWR:** A thrust-to-weight ratio of at least 1.5:1 to support post-stall pivoting and rapid energy recovery.
- **Sensor suite:** Triple-redundant IMUs, GNSS, synthetic air-data system (SADS), and structural/thermal sensors feeding Layer 2.

0.2.3 Control Interface

The airframe accepts only the normalized rigid-body command vector:

$$\mathbf{u}_{RB} = (p, q, r, T, B)$$

as stated explicitly in the Top-Level Specification [?]. No tactical or latent representations reach the airframe; all higher-level logic is upstream in Layers 2 and 3.

0.3 Three-Layer Control Architecture

The SRBM control system is organized into three layers with strict authority boundaries. This structure is described in both the Top-Level Specification and the Unified Tactical Architecture documents [?, ?].

0.3.1 Layer 3 — Tactical Autonomy

Layer 3 generates maneuver intent based on:

- fused sensor state,
- doctrine parameters (α, γ) ,

- tactical anchor behaviors (HO, HD, DO, DD),
- engagement geometry and mission context.

It outputs only the normalized rigid-body command vector:

$$(p_{cmd}, q_{cmd}, r_{cmd}, T_{cmd}, B_{cmd})$$

as confirmed in the Top-Level Specification [?].

0.3.2 Layer 2 — Envelope Enforcement

Layer 2 enforces:

- structural limits (load factor n , rate $\dot{n}$),
- aerodynamic limits (AoA, sideslip),
- dynamic pressure constraints,
- mission-level constraints (keep-out zones, collision avoidance),
- actuator feasibility and rate limits.

It reshapes unsafe commands into:

$$\mathbf{u}_{RB,safe}$$

as described in the Unified Tactical Architecture [?].

0.3.3 Layer 1 — Moment Allocation

Layer 1 converts the safe rigid-body command into actuator deflections via the control-effectiveness matrix B_x :

$$\mathbf{u}_{act} = B_x^\dagger \mathbf{r}_{mom}$$

where $\mathbf{r}_{mom}$ is the desired moment vector derived from $\mathbf{u}_{RB,safe}$.

This layer:

- enforces symmetry,
- handles redundant surfaces,

- applies deflection and rate limits,
- commands all hardware actuators.

Layer 1 is the only layer with direct hardware authority [?].

Part II

Tactical Architecture

0.4 Unified Doctrine Model

The Unified Doctrine Model defines the continuous tactical space used by Layer 3 to generate maneuver intent. As described in the Unified Tactical Architecture document, tactical behavior is parameterized by two continuous axes—coordination α and combat posture γ [?].

0.4.1 Doctrine Axes

- **Coordination axis α :** Hive $\leftrightarrow$ Individual Represents the degree of swarm coupling. High α produces coordinated, multi-aircraft geometries; low α produces individualistic maneuvering.
- **Combat posture axis γ :** Defensive $\leftrightarrow$ Offensive Derived from force advantage, energy advantage, sensor advantage, and effective health [?].

Both α and γ are continuous, bounded in $[0, 1]$, and temporally smoothed.

0.4.2 State Inputs to Doctrine

The posture axis γ is computed from normalized state variables:

$$\gamma_{\text{raw}} = c_{\beta}\beta + c_E E + c_S S + c_H H_{\text{eff}}$$

with logistic shaping:

$$\gamma = \frac{1}{1 + e^{-k(\gamma_{\text{raw}} - 0.5)}}$$

and temporal filtering:

$$\dot{\gamma} = \frac{\gamma_{\text{target}} - \gamma}{\tau}$$

These definitions follow the Unified Tactical Architecture [?].

0.4.3 Doctrine Output

The doctrine layer outputs:

$$D = \{\alpha, \gamma, w_{HO}, w_{HD}, w_{DO}, w_{DD}, u_{HO}, u_{HD}, u_{DO}, u_{DD}\}$$

where u_i are anchor tactical vectors and w_i are convex weights.

0.5 Tactical Anchor Behaviors

Four doctrinal anchor behaviors span the tactical space [?]:

$$i \in \{HO, HD, DO, DD\}$$

Each anchor produces a normalized tactical vector:

$$u_i = (u_{x,i}, u_{y,i}, u_{z,i}, u_{T,i})$$

These represent desired roll, pitch, yaw/LOS-rate shaping, and thrust/energy bias.

0.5.1 Hive–Offensive (HO)

Coordinated multi-axis offensive pressure:

- Encirclement and distributed headings
- LOS-rate pressure around the target
- Energy-optimal attack geometry

0.5.2 Hive–Defensive (HD)

Coordinated survivability and spacing:

- Move toward swarm centroid or escape corridor
- Pitch toward energy-positive escape attitude
- LOS-rate shaping away from threat
- Thrust bias toward energy recovery

0.5.3 Dogfight–Offensive (DO)

Individual angle generation:

- Bank toward optimal turn-rate bank
- Pitch toward lead/pure pursuit
- Yaw/sideslip alignment for nose-on geometry
- Thrust bias for fight-sustaining energy

0.5.4 Dogfight–Defensive (DD)

Survival and overshoot forcing:

- Break-bank toward max instantaneous turn
- Jink pitch for vertical unpredictability
- LOS-rate disruption to break tracking
- Thrust bias toward escape energy band

0.6 Convex Tactical Blending

The doctrine layer computes convex weights:

$$w_i \geq 0, \quad \sum_i w_i = 1$$

using the biquadratic interpolation rule defined in the Unified Tactical Architecture [?]:

$$\begin{aligned} w_{HO} &= \alpha\gamma \\ w_{HD} &= \alpha(1 - \gamma) \\ w_{DO} &= (1 - \alpha)\gamma \\ w_{DD} &= (1 - \alpha)(1 - \gamma) \end{aligned}$$

0.6.1 Blended Tactical Intent

The blended tactical vector is:

$$u_{\text{tac}} = \sum_{i \in \{HO, HD, DO, DD\}} w_i u_i$$

This vector remains internal to Layer 3 and is not exposed externally.

0.6.2 Frame Consistency

If anchor vectors are expressed in world coordinates:

$$u_{\text{tac}}^b = R_{WB} u_{\text{tac}}^w$$

ensuring orientation-agnostic behavior [?].

0.7 Latent Tactical Intent Space

Appendix C defines the internal latent representation used by Layer 3 [?]. Each anchor behavior produces a latent vector:

$$z_i = (z_{i,1}, z_{i,2}, \dots, z_{i,d})$$

where each component is bounded:

$$-1 \leq z_{i,k} \leq 1$$

0.7.1 Latent Blending

The supervisor computes weights w_i and blends latent vectors componentwise:

$$z_{\text{tac},k} = w_{HO}z_{HO,k} + w_{HD}z_{HD,k} + w_{DO}z_{DO,k} + w_{DD}z_{DD,k}$$

In vector form:

$$z_{\text{tac}} = \sum_i w_i z_i$$

0.7.2 RigidBody Decoder

Layer 3 converts the blended latent vector into the rigid-body command:

$$(p_{cmd}, q_{cmd}, r_{cmd}, T_{cmd}, B_{cmd})$$

via shared decoder functions:

$$p_{cmd} = f_p(z_{\text{tac}}), \quad q_{cmd} = f_q(z_{\text{tac}}), \quad r_{cmd} = f_r(z_{\text{tac}})$$

$$T_{cmd} = f_T(z_{\text{tac}}), \quad B_{cmd} = f_B(z_{\text{tac}})$$

These functions enforce smoothness, boundedness, and Layer 2 compatibility [?].

0.7.3 Layer Boundaries

Layer 3 outputs only the rigid-body command vector. Latent vectors and blending weights are never exposed to Layer 2 or Layer 1.

Part III

Intent-to-RigidBody Mapping

0.8 Layer 3 Intent-to-RigidBody Mapping Pipeline

Layer 3 converts blended tactical intent into the normalized rigid-body command vector:

$$(p, q, r, T, B)$$

as described in the Intent-to-RigidBody Mapping Specification [?]. This mapping is nonlinear, state-dependent, and designed to be compatible with Layer 2 envelope enforcement.

0.8.1 Inputs

Layer 3 receives:

- blended tactical intent u_{tac}^w (world frame),
- anchor weights w_i ,
- aircraft state (quaternion q_{WB} , velocity, AoA, sideslip),
- structural and energy state,
- doctrine context (HO/HD/DO/DD).

0.8.2 Step 1 — Doctrine Blending

The blended tactical intent is computed as:

$$u_{\text{tac}}^w = \sum_{i \in \{HO, HD, DO, DD\}} w_i u_i^w$$

as defined in the Unified Tactical Architecture [?].

0.8.3 Step 2 — World-to-Body Rotation

Angular components are rotated into body axes:

$$u_{\text{tac}}^b = R_{WB} u_{\text{tac}}^w$$

ensuring orientation-agnostic behavior [?].

0.8.4 Step 3 — Smooth Saturation

Each component is passed through a smooth, odd, bounded nonlinearity:

$$\sigma(u) = \tanh(ku)$$

producing raw commands:

$$p_{\text{raw}}, q_{\text{raw}}, r_{\text{raw}}, T_{\text{raw}}, B_{\text{raw}}$$

0.8.5 Step 4 — Authority Envelope Scaling

Authority fractions G_p, G_q, G_r, G_T, G_B are predicted by envelope networks using the full state vector x [?].

0.8.6 Step 5 — Final RigidBody Command

Signed angular channels:

$$p = G_p p_{\text{raw}}, \quad q = G_q q_{\text{raw}}, \quad r = G_r r_{\text{raw}}$$

Nonnegative energy channels:

$$T = G_T T_{\text{raw}}, \quad B = G_B B_{\text{raw}}$$

0.8.7 Output

Layer 3 outputs only:

$$(p, q, r, T, B)$$

Latent vectors, anchor weights, and tactical intent remain internal to Layer 3 and are not exposed to Layer 2 or Layer 1 [?].

0.9 Smooth Saturation and Nonlinear Shaping

Layer 3 applies smooth saturation functions to ensure all intermediate commands remain bounded, continuous, and Layer 2 compatible.

0.9.1 Saturation Function

The canonical shaping function is:

$$\sigma(u) = \tanh(ku)$$

where k controls aggressiveness.

Properties:

- odd symmetry,
- smooth and differentiable,
- bounded in $[-1, 1]$,
- no discontinuities,
- no impulsive transitions when doctrine weights shift.

0.9.2 Angular Channels

Angular intent components are shaped as:

$$p_{\text{raw}} = \sigma_p(u_{x,b}), \quad q_{\text{raw}} = \sigma_q(u_{y,b}), \quad r_{\text{raw}} = \sigma_r(u_{z,b})$$

0.9.3 Energy Channels

Energy intent is shaped into thrust and braking:

$$T_{\text{raw}} = \sigma_T(u_T), \quad B_{\text{raw}} = \sigma_B(-u_T)$$

This ensures:

- thrust and brake remain nonnegative,
- $T_{\text{raw}}, B_{\text{raw}} \in [0, 1]$,
- energy commands are smooth and compatible with Layer 2.

0.9.4 Continuity Guarantees

As stated in the Intent-to-RigidBody Mapping Specification [?], Layer 3 guarantees:

- no discontinuities across doctrine transitions,

- no sign inversions,
- no impulsive changes in p, q, r, T, B ,
- consistent body-frame alignment.

0.10 Learned Authority Envelopes

Layer 3 uses learned authority envelopes to ensure all commands remain compatible with Layer 2's structural and aerodynamic limits. These envelopes are learned directly from Layer 2 behavior [?].

0.10.1 Envelope Networks

Each channel has a neural network predicting the safe authority fraction:

$$G_p(x), G_q(x), G_r(x), G_T(x), G_B(x) \in [0, 1]$$

Inputs include:

- dynamic pressure q_d ,
- AoA and sideslip,
- load factor n and rate $\dot{n}$,
- body-frame velocity,
- energy state,
- anchor encoding.

0.10.2 Final Command Computation

Signed angular channels:

$$p = G_p p_{\text{raw}}, \quad q = G_q q_{\text{raw}}, \quad r = G_r r_{\text{raw}}$$

Energy channels:

$$T = G_T T_{\text{raw}}, \quad B = G_B B_{\text{raw}}$$

0.10.3 Learning Objective

For each channel, Layer 3 observes:

$$c_{\text{prop}}, \quad c_{\text{L2}}$$

and computes the target authority fraction:

$$y = \text{clip}\left(\frac{c_{\text{L2}}}{\max(c_{\text{prop}}, \epsilon)}, 0, 1\right)$$

The networks converge to the true safe authority surface enforced by Layer 2.

0.10.4 Behavioral Guarantees

As stated in the mapping specification [?]:

- $p, q, r \in [-1, 1]$,
- $T, B \in [0, 1]$,
- smoothness ensured by σ functions and NNs,
- doctrine-aware scaling,
- state-aware authority limits,
- aircraft-agnostic behavior.

Part IV

Envelope Enforcement

0.11 Structural Load Enforcement

Layer 2 enforces structural limits on load factor, load rate, and dynamic pressure. These limits are authoritative and override all upstream commands. The Unified Tactical Architecture describes Layer 2 as the “structural and aerodynamic safety boundary” for SRBM [?].

0.11.1 Load Factor Limit

The structural load factor n is constrained:

$$|n| \leq n_{\max}$$

with a rate constraint:

$$|\dot{n}| \leq \dot{n}_{\max}$$

If the commanded rigid-body vector (p, q, r, T, B) would exceed these limits, Layer 2 scales angular channels:

$$(p, q, r) \leftarrow s_n(p, q, r)$$

where $s_n \in [0, 1]$ is the minimum scaling factor required to satisfy both n and $\dot{n}$ constraints.

0.11.2 Dynamic Pressure Limit

Dynamic pressure is constrained:

$$q_d \leq q_{d,\max}$$

If violated, Layer 2 reduces thrust and angular authority:

$$T \leftarrow s_q T, \quad (p, q, r) \leftarrow s_q(p, q, r)$$

where s_q is computed from the margin to $q_{d,\max}$.

0.11.3 Structural Health Integration

Structural health H_{phys} modulates limits:

$$n_{\max}(H_{phys}), \quad \dot{n}_{\max}(H_{phys})$$

As health degrades, allowable loads decrease, ensuring survivability.

0.12 Aerodynamic Envelope Enforcement

Layer 2 enforces aerodynamic limits on angle of attack (AoA), sideslip, and post-stall behavior. These limits ensure the aircraft remains within the controllable aerodynamic regime.

0.12.1 Angle of Attack Limit

AoA is constrained:

$$|\text{AoA}| \leq \text{AoA}_{\max}$$

If the commanded pitch rate q would increase AoA beyond the limit:

$$q \leftarrow s_\alpha q$$

where s_α is computed from the predicted AoA derivative.

0.12.2 Sideslip Limit

Sideslip is constrained:

$$|\beta| \leq \beta_{\max}$$

If yaw command r would increase $|\beta|$:

$$r \leftarrow s_\beta r$$

0.12.3 Post-Stall Behavior

SRBM airframes are designed for full-attitude maneuvering, but Layer 2 still enforces:

- stall margin protection,
- smooth transitions through high-AoA regimes,
- damping augmentation when aerodynamic derivatives weaken.

These behaviors are described in the Unified Tactical Architecture [?].

0.13 Rate Shaping and Feasibility Enforcement

Layer 2 ensures that all rigid-body commands are dynamically feasible and compatible with actuator bandwidth.

0.13.1 Angular Rate Limits

Angular channels are constrained:

$$|p| \leq p_{\max}, \quad |q| \leq q_{\max}, \quad |r| \leq r_{\max}$$

If exceeded:

$$(p, q, r) \leftarrow s_{\text{rate}}(p, q, r)$$

0.13.2 Command Slew Limits

Layer 2 enforces first-order rate limits:

$$\dot{p} \leq \dot{p}_{\max}, \quad \dot{q} \leq \dot{q}_{\max}, \quad \dot{r} \leq \dot{r}_{\max}$$

This prevents impulsive transitions and ensures compatibility with Layer 1 actuator limits.

0.13.3 Energy Channel Shaping

Thrust and brake channels are shaped:

$$T \leftarrow \min(T, T_{\max}), \quad B \leftarrow \min(B, B_{\max})$$

with rate limits:

$$\dot{T} \leq \dot{T}_{\max}, \quad \dot{B} \leq \dot{B}_{\max}$$

0.13.4 Feasibility Projection

If any constraint is violated, Layer 2 computes a feasibility projection:

$$\mathbf{u}_{RB, safe} = \Pi_{\mathcal{E}}(\mathbf{u}_{RB})$$

where $\mathcal{E}$ is the feasible envelope defined by structural, aerodynamic, and rate limits.

0.14 Mission-Level Constraints

Layer 2 also enforces mission-level constraints that ensure tactical behavior remains safe and coordinated.

0.14.1 Collision Avoidance

Minimum separation distance:

$$R_{ij} \geq R_{\min}$$

If violated, Layer 2 injects a repulsive correction into (p, q, r) .

0.14.2 Keep-Out Zones

Geofenced regions or threat volumes are enforced by projecting commands away from forbidden sets.

0.14.3 Energy Floor

A minimum energy level is enforced:

$$E \geq E_{\min}$$

If violated:

$$T \leftarrow 1, \quad B \leftarrow 0$$

forcing an energy recovery maneuver.

0.14.4 Thermal and Structural Health

Thermal load and structural health H_{phys} modulate allowable limits and may trigger:

- thrust reduction,
- angular authority reduction,
- emergency escape behaviors.

Part V

Moment Allocation and Actuation

0.15 Rigid-Body Moment Mapping

Layer 1 converts the envelope-safe rigid-body command vector

$$\mathbf{u}_{RB,safe} = (p, q, r, T, B)$$

into a desired moment vector

$$\mathbf{r}_{mom} = (M_x, M_y, M_z)$$

as described in the Intent-to-RigidBody Mapping Specification [?].

0.15.1 Angular Channels

Angular commands are mapped to desired body-axis moments using state-dependent gains:

$$M_x = K_p p, \quad M_y = K_q q, \quad M_z = K_r r$$

The gains K_p, K_q, K_r depend on:

- dynamic pressure q_d ,
- airspeed,
- actuator authority,
- structural health.

0.15.2 Energy Channels

Thrust and braking commands are passed directly to propulsion and braking subsystems:

$$T_{cmd} = T, \quad B_{cmd} = B$$

These channels bypass the moment allocator but remain subject to Layer 2 rate and feasibility limits.

0.16 Control Effectiveness Matrix

The control effectiveness matrix B_x maps actuator deflections to body-axis moments:

$$\mathbf{r}_{mom} = B_x \mathbf{u}_{act} \tag{1}$$

Each column of B_x corresponds to an actuator, and each row corresponds to a body-axis moment.

0.16.1 State Dependence

As described in the Unified Tactical Architecture, B_x depends on:

- dynamic pressure q_d ,
- angle of attack and sideslip,
- Mach number,
- actuator configuration,
- structural health.

0.16.2 Symmetric Airframe Structure

For SRBM's symmetric airframe:

- left/right actuators contribute equally with opposite signs,
- dorsal/ventral actuators contribute equally with opposite signs,
- no privileged orientation exists.

This symmetry simplifies the structure of B_x and improves conditioning.

0.16.3 Control Effectiveness Degradation

When structural or aerodynamic health degrades, B_x is modified:

$$B_x \leftarrow H_{phys} B_x \tag{2}$$

ensuring safe operation even with partial actuator loss.

0.17 Moment Allocation Solver

Layer 1 solves the constrained least squares allocation problem:

$$\mathbf{u}_{act} = \arg \min_{\mathbf{u}} \|B_x \mathbf{u} - \mathbf{r}_{mom}\|^2 \tag{3}$$

subject to actuator limits.

0.17.1 Minimum Norm Solution

The unconstrained solution is the Moore–Penrose pseudoinverse:

$$\mathbf{u}_{act} = B_x^\dagger \mathbf{r}_{mom} \quad (4)$$

This is the minimum norm actuator vector that achieves the desired moment.

0.17.2 Null Space Shaping

When the system is redundant, the null space is used for symmetry, drag minimization, thermal balancing, and wear leveling.

The general solution is:

$$\mathbf{u}_{act} = B_x^\dagger \mathbf{r}_{mom} + \left(I - B_x^\dagger B_x\right) \mathbf{u}_{null} \quad (5)$$

0.17.3 Actuator Limits

Each actuator obeys:

$$\delta_{min} \leq \delta_i \leq \delta_{max} \quad (6)$$

and

$$\left|\dot{\delta}_i\right| \leq \dot{\delta}_{max} \quad (7)$$

If any limit is violated, Layer 1 performs a feasibility projection:

$$\mathbf{u}_{act} \leftarrow \Pi_{\mathcal{A}}(\mathbf{u}_{act}) \quad (8)$$

where $\Pi_{\mathcal{A}}$ is the projection operator onto the convex actuator constraint set $\mathcal{A}$.

0.17.4 Final Output

Layer 1 outputs the actuator command vector:

$$\mathbf{u}_{act} = (\delta_1, \delta_2, \dots, \delta_N) \quad (9)$$

which is the only layer with direct hardware authority.

Part VI

Tactical Cost Functions and RL Training

0.18 HiveOffensive Cost Function

The HiveOffensive (HO) behavior represents coordinated multi-aircraft offensive pressure. The cost function is defined in the Tactical Cost Specification and Appendix A.

0.18.1 Cost Definition

The HO cost encourages:

- reducing range to the target,
- increasing closure rate,
- maintaining offensive LOS-rate pressure,
- maximizing energy advantage.

The cost is:

$$J_{HO} = w_R R_{ij} + w_{\dot{R}} \dot{R}_{ij} + w_{LOS} \|\dot{L}\dot{O}S\| + w_E(1 - E)$$

All weights are positive and behavior-specific.

0.19 HiveDefensive Cost Function

The HiveDefensive (HD) behavior represents coordinated survivability, spacing, and escape geometry.

0.19.1 Cost Definition

The HD cost encourages:

- increasing spacing from threats,
- maximizing energy margin,
- preserving effective health,
- maintaining sensor advantage.

The cost is:

$$J_{HD} = w_R \frac{1}{R_{ij}} + w_E(1 - E) + w_H(1 - H_{\text{eff}}) + w_S(1 - S)$$

All weights are positive and tuned for defensive coordination.

0.20 DogfightOffensive Reward Function

The DogfightOffensive (DO) behavior is trained using reinforcement learning. The reward is the negative of the DO cost defined in Appendix A.

0.20.1 Reward Definition

$$R_{DO} = -J_{DO}$$

where:

$$J_{DO} = w_{\text{LOS}} \|\dot{\text{LOS}}\| + w_R R_{ij} + w_{\dot{R}} \dot{R}_{ij} + w_E (1 - E)$$

0.20.2 Shaping Terms

Additional shaping improves learning stability:

$$R_E = k_E (E - E_{\text{target}})$$

$$R_{\text{LOS}} = -k_{\text{LOS}} \|\dot{\text{LOS}}\|$$

$$R_R = -k_R R_{ij}$$

The final reward is the sum of base and shaping terms.

0.21 DogfightDefensive Reward Function

The DogfightDefensive (DD) behavior is trained using reinforcement learning. The reward is the negative of the DD cost defined in Appendix A.

0.21.1 Reward Definition

$$R_{DD} = -J_{DD}$$

where:

$$J_{DD} = w_{\text{LOS}} \frac{1}{\|\dot{\text{LOS}}\|} + w_R \frac{1}{R_{ij}} + w_E (1 - E) + w_H (1 - H_{\text{eff}})$$

0.21.2 Shaping Terms

$$R_H = k_H H_{\text{eff}}$$

$$R_E = k_E (E - E_{\text{target}})$$

Terminal events (kill, death, collision, out-of-bounds) apply large terminal rewards.

0.22 Reinforcement Learning Curriculum

The RL curriculum trains DO and DD behaviors across progressively more difficult engagement scenarios.

0.22.1 Curriculum Stages

1. **Kinematic pursuit/escape** No aerodynamics; pure geometry.
2. **Low-AoA maneuvering** Aerodynamic model enabled; no stalls.
3. **Full envelope maneuvering** High-AoA, post-stall, and energy dynamics enabled.
4. **Adversarial training** Opponent uses scripted or learned policies.
5. **Domain randomization** Mass, inertia, thrust, sensor noise randomized.

0.22.2 Policy Freezing

Once trained, DO and DD policies are frozen and used as anchor behaviors in the doctrine model.

0.22.3 Compatibility with Layer 2

All RL actions are passed through Layer 2 envelope enforcement, ensuring RL cannot command unsafe behavior.

.1 Tactical Cost and Reward Function Definitions

This appendix provides the authoritative mathematical definitions for the tactical cost functions used by the four doctrinal behaviors: HiveOffensive (HO), HiveDefensive (HD), DogfightOffensive (DO), and DogfightDefensive (DD). These definitions are derived from the SRBM Tactical Cost Specification.

.1.1 A.1 Common Engagement Quantities

All tactical behaviors use the following common quantities:

- relative position vector: $\mathbf{r}_{ij}$,
- range: $R_{ij} = \|\mathbf{r}_{ij}\|$,
- closure rate: $\dot{R}_{ij}$,
- line-of-sight rate: $\dot{\text{LOS}}$,
- specific energy: E_s ,
- normalized energy advantage: E ,
- sensor advantage: S ,
- effective health: H_{eff} .

All weights w . are positive and behavior-specific.

.1.2 A.2 HiveOffensive (HO) Cost Function

The HO behavior encourages coordinated offensive pressure, encirclement, and energy advantage.

$$J_{HO} = w_R R_{ij} + w_{\dot{R}} \dot{R}_{ij} + w_{\text{LOS}} \|\dot{\text{LOS}}\| + w_E (1 - E) \quad (10)$$

Lower cost corresponds to stronger offensive posture.

.1.3 A.3 HiveDefensive (HD) Cost Function

The HD behavior encourages coordinated spacing, survivability, and energy preservation.

$$J_{HD} = w_R \frac{1}{R_{ij}} + w_E (1 - E) + w_H (1 - H_{\text{eff}}) + w_S (1 - S) \quad (11)$$

Lower cost corresponds to safer defensive geometry.

.1.4 A.4 DogfightOffensive (DO) Cost Function

The DO behavior is trained via reinforcement learning. Its cost function encourages angle generation, pursuit geometry, and energy advantage.

$$J_{DO} = w_{\text{LOS}} \|\dot{\text{LOS}}\| + w_R R_{ij} + w_{\dot{R}} \dot{R}_{ij} + w_E (1 - E) \quad (12)$$

The RL reward is the negative of the cost:

$$R_{DO} = -J_{DO} \quad (13)$$

.1.5 A.5 DogfightDefensive (DD) Cost Function

The DD behavior is also RL-trained. Its cost encourages overshoot forcing, escape geometry, and health preservation.

$$J_{DD} = w_{\text{LOS}} \frac{1}{\|\dot{\text{LOS}}\|} + w_R \frac{1}{R_{ij}} + w_E(1 - E) + w_H(1 - H_{\text{eff}}) \quad (14)$$

The RL reward is:

$$R_{DD} = -J_{DD} \quad (15)$$

.1.6 A.6 Shaping Terms for RL

To improve learning stability, shaping terms are added:

$$R_E = k_E(E - E_{\text{target}}) \quad (16)$$

$$R_{\text{LOS}} = -k_{\text{LOS}}\|\dot{\text{LOS}}\| \quad (17)$$

$$R_R = -k_R R_{ij} \quad (18)$$

$$R_H = k_H H_{\text{eff}} \quad (19)$$

These terms do not change the optimal policy but accelerate convergence.

.1.7 A.7 Terminal Rewards

Terminal events apply large-magnitude rewards:

- kill or capture geometry,
- being killed,
- collision,

- out-of-bounds,
- energy depletion.

Terminal rewards dominate shaping and enforce correct tactical behavior.

.1.8 A.8 Summary

This appendix defines the complete mathematical structure of the SRBM tactical cost and reward framework. These definitions are used directly in Part VI for HO/HD analytic behaviors and DO/DD reinforcement learning.

.2 Observation Tensor Specification

Layer 3 receives a fixed-size observation tensor O_t every 10–20 ms. The tensor is designed to expose only the low-frequency, rigid-body quantities required for tactical decision making. All high-frequency aerodynamic disturbances, actuator nonlinearities, and transient perturbations are absorbed by the high-rate control substrate (Layers 1–2), ensuring that the tactical layer interacts only with a clean, stable, and airframe-agnostic representation of the vehicle state.

A core architectural principle of SRBM is that the physical airframe must present itself to Layer 3 as an idealized, symmetric, 6-DoF rigid-body entity. The observation tensor is the mechanism by which this abstraction is enforced. In particular, the execution-gap block O_t^{gap} serves as the formal interface between physics and cognition: it provides a filtered, bounded measure of how well the virtualization substrate achieved the commanded rigid-body motion during the previous tactical epoch.

The tensor is:

- frame-agnostic,
- rigid-body native,
- airframe-agnostic,
- composed only of physically meaningful quantities,
- compatible with all HO/HD/DO/DD cost terms.

The tensor is divided into four blocks:

$$O_t = \{O_t^{\text{self}}, O_t^{\text{gap}}, O_t^{\text{hive}}, O_t^{\text{threat}}\} \quad (20)$$

.2.1 B.1 Self-State Block O_t^{self}

This block encodes the instantaneous rigid-body state of the aircraft.

B.1.1 Orientation (Quaternion)

$$(q_w, q_x, q_y, q_z) \quad (21)$$

B.1.2 Body Angular Rates

$$(p, q, r) \quad (22)$$

B.1.3 Body-Frame Velocity

$$(v_x, v_y, v_z) \quad (23)$$

B.1.4 Body-Frame Acceleration

$$(a_x, a_y, a_z) \quad (24)$$

B.1.5 Energy State

$$(E_s, P_s) \quad (25)$$

B.1.6 Control Surface and TVC State

$$(\delta_{r1}, \delta_{r2}, \delta_{r3}, \delta_{r4}, \delta_{\text{TVC}}) \quad (26)$$

.2.2 B.2 Execution-Gap Block O_t^{gap}

This block encodes the difference between commanded and realized rigid-body motion from the previous timestep. It is the primary abstraction boundary between the physical plant and tactical cognition.

Rather than exposing high-frequency disturbances, O_t^{gap} provides a low-pass, envelope-filtered measure of:

- Layer 2 clamping,
- actuator saturation,
- structural or aerodynamic constraint enforcement,
- control authority shortfall.

This allows Layer 3 to adapt to the *behavior* of the virtualization substrate, not to raw turbulence or actuator noise.

B.2.1 Commanded Rigid-Body Vector (at $t - 1$)

$$(p_{\text{cmd}}, q_{\text{cmd}}, r_{\text{cmd}}, T_{\text{cmd}}, B_{\text{cmd}}) \quad (27)$$

B.2.2 Executed Rigid-Body Vector (at $t - 1$)

$$(p_{\text{exec}}, q_{\text{exec}}, r_{\text{exec}}, T_{\text{exec}}, B_{\text{exec}}) \quad (28)$$

B.2.3 Execution-Gap Components

$$\begin{aligned} r_{\text{gap}}^p &= p_{\text{cmd}} - p_{\text{exec}} \\ r_{\text{gap}}^q &= q_{\text{cmd}} - q_{\text{exec}} \\ r_{\text{gap}}^r &= r_{\text{cmd}} - r_{\text{exec}} \\ r_{\text{gap}}^T &= T_{\text{cmd}} - T_{\text{exec}} \\ r_{\text{gap}}^B &= B_{\text{cmd}} - B_{\text{exec}} \end{aligned} \quad (29)$$

B.2.4 Observation Block

$$O_t^{\text{gap}} = (r_{\text{gap}}^p, r_{\text{gap}}^q, r_{\text{gap}}^r, r_{\text{gap}}^T, r_{\text{gap}}^B) \quad (30)$$

.2.3 B.3 Hive / Neighbor Block O_t^{hive}

This block encodes cooperative swarm geometry and status for up to N_{max} neighbors.

B.3.1 Relative Positions (Body Frame)

$$(\Delta p_{x,k}, \Delta p_{y,k}, \Delta p_{z,k}) \quad (31)$$

B.3.2 Relative Orientations (Relative Quaternions)

$$(\Delta q_{w,k}, \Delta q_{x,k}, \Delta q_{y,k}, \Delta q_{z,k}) \quad (32)$$

B.3.3 Neighbor Status Flags

$$(s_k^{\text{ammo}}, s_k^{\text{fuel}}, s_k^{\text{engaged}}) \quad (33)$$

.2.4 B.4 Threat / Target Block O_t^{threat}

This block encodes the perceived state of up to M_{max} threats.

B.4.1 LOS Vector (Body Frame)

$$(r_{x,j}^b, r_{y,j}^b, r_{z,j}^b) \quad (34)$$

B.4.2 Enemy Velocity Direction (Body Frame)

$$(v_{ex,j}^b, v_{ey,j}^b, v_{ez,j}^b) \quad (35)$$

B.4.3 Closure Rate

$$V_{c,j} = \dot{R}_{ij} \quad (36)$$

B.4.4 Threat Exposure Scalar

$$E_j \quad (37)$$

.2.5 B.5 Masking and Padding

Both the hive and threat blocks include:

- a fixed number of slots,
- a mask vector indicating which slots contain valid data.

This ensures:

- constant tensor size,
- deterministic neural-network input,
- stable behavior under varying swarm sizes.

.2.6 B.6 Observation Tensor Summary

The complete tensor is:

- frame-agnostic,
- rigid-body native,
- aircraft-agnostic,
- updated every 10–20 ms,

- sufficient for high-agility, post-stall, TVC-enabled maneuvering.

This appendix constitutes the authoritative definition of the Layer 3 observation tensor.

.3 Latent Tactical Intent Space

This appendix defines the internal latent tactical representation used by Layer 3. Each anchor behavior produces a latent vector that encodes its tactical preferences in a geometry-native space. These latent vectors represent abstract tactical directions rather than rigid-body axes or control-surface deflections. Layer 3 blends these vectors and decodes the result into the normalized rigid-body command vector $(p_{cmd}, q_{cmd}, r_{cmd}, T_{cmd}, B_{cmd})$.

The latent space is the cognitive manifold of SRBM: it expresses high-level tactical intent in a form that is independent of airframe geometry, aerodynamic assumptions, or actuator configuration. Only after blending is this intent mapped into the rigid-body command space through a shared decoder.

.3.1 C.1 Behavior-Specific Latent Vectors

Each tactical behavior satisfies:

$$i \in \{HO, HD, DO, DD\} \quad (38)$$

and produces a latent vector:

$$z_i = (z_{i,1}, z_{i,2}, \dots, z_{i,d}) \quad (39)$$

Properties:

- d is the latent dimensionality.
- Each component is bounded:

$$-1 \leq z_{i,k} \leq 1 \quad (40)$$

Components encode abstract tactical preferences such as:

- increase LOS rate,
- maintain encirclement,
- open range,

- maximize snapshot opportunity,
- break lock,
- preserve energy.

These components do not correspond to physical axes. They represent geometry-native tactical directions that remain valid across all airframes compatible with SRBM.

.3.2 C.2 Supervisor-Generated Behavior Weights

The supervisor computes convex weights:

$$w_{HO}, w_{HD}, w_{DO}, w_{DD} \quad (41)$$

Weights satisfy:

$$w_{HO} + w_{HD} + w_{DO} + w_{DD} = 1 \quad (42)$$

These weights determine how strongly each behavior influences the final latent tactical intent.

.3.3 C.3 Blended Latent Tactical Vector

The blended latent vector is computed componentwise:

$$z_{\text{tac},k} = w_{HO}z_{HO,k} + w_{HD}z_{HD,k} + w_{DO}z_{DO,k} + w_{DD}z_{DD,k} \quad (43)$$

In vector form:

$$z_{\text{tac}} = (z_{\text{tac},1}, z_{\text{tac},2}, \dots, z_{\text{tac},d}) \quad (44)$$

This representation ensures smooth transitions between behaviors and prevents conflicting rigid-body commands. Blending occurs entirely in the latent space, before any mapping to physical motion.

.3.4 C.4 Shared RigidBody Decoder (Layer 3 to Layer 2)

Layer 3 converts the blended latent vector into the normalized rigid-body command vector. The decoder is the sole interface between tactical cognition and the rigid-body command space.

C.4.1 Decoder Output

$$(p_{\text{cmd}}, q_{\text{cmd}}, r_{\text{cmd}}, T_{\text{cmd}}, B_{\text{cmd}}) \quad (45)$$

C.4.2 Decoder Functions

Each commanded component is produced by a decoder function:

$$p_{cmd} = f_p(z_{tac}), \quad q_{cmd} = f_q(z_{tac}), \quad r_{cmd} = f_r(z_{tac}) \quad (46)$$

$$T_{cmd} = f_T(z_{tac}), \quad B_{cmd} = f_B(z_{tac}) \quad (47)$$

Decoder properties:

- shared across all behaviors,
- smooth and Lipschitz-continuous,
- bounded outputs,
- independent of airframe geometry,
- compatible with Layer 2 envelope enforcement.

Layer 2 receives only these five values. Latent vectors and weights remain internal to Layer 3 and are never exposed externally.

.3.5 C.5 Latent Space Constraints

C.5.1 Bounded Values

$$-1 \leq z_{i,k} \leq 1 \quad (48)$$

C.5.2 Smoothness

Decoder functions are designed or trained to be:

- Lipschitz-continuous,
- free of discontinuities,
- robust to unseen mixtures of behaviors.

C.5.3 Airframe-Agnostic

The latent space is geometric, not aerodynamic. Layer 1 handles all actuator mapping and control-effectiveness considerations.

.3.6 C.6 Advantages of the Latent Tactical Architecture

C.6.1 Smooth Behavior Blending

Weighted latent vectors blend seamlessly, avoiding:

- command conflicts,
- discontinuous transitions,
- mode-switch oscillations.

C.6.2 Emergent Hybrid Tactics

The decoder interprets latent mixtures, enabling:

- offensive–defensive hybrids,
- cooperative–individual blends,
- energy–maneuvering tradeoffs.

C.6.3 Safety and Robustness

The decoder enforces:

- rigid-body limits,
- smoothness,
- bounded outputs,
- independence from tactical mode.

C.6.4 Airframe Independence

The latent space works for:

- symmetric airframes,
- thrust-vectoring vehicles,
- high-AoA and post-stall regimes.

.3.7 C.7 Appendix Status

Appendix C defines the canonical latent tactical intent architecture used internally by Layer 3. Layer 3 outputs only the rigid-body command vector $(p_{cmd}, q_{cmd}, r_{cmd}, T_{cmd}, B_{cmd})$ to Layer 2. Latent vectors and blending weights are never exposed externally.

.4 Example SRBM-Compliant Airframe

This appendix provides a reference airframe that is fully compliant with the SRBM architecture. The purpose is illustrative only: SRBM remains airframe-agnostic, and any vehicle providing a valid control-effectiveness matrix B_x and actuator constraint set may be used.

A core architectural requirement of SRBM is that the physical airframe present itself to Layer 3 as an idealized, symmetric, 6-DoF rigid-body entity. Layers 1 and 2 form the virtualization substrate that guarantees this abstraction by absorbing aerodynamic nonlinearities, actuator limits, and transient disturbances. The example airframe below demonstrates one possible instantiation of this requirement.

.4.1 D.1 Airframe Overview

The reference vehicle is a symmetric, VTOL-capable, cockpitless rigid body with:

- left/right symmetry,
- dorsal/ventral symmetry,
- zero dihedral,
- no built-in lift or stability bias,
- no privileged orientation.

The airframe operates as a pure 6-DoF rigid body. All lift, stability, and maneuvering authority are generated through commanded attitude, bidirectional camber, thrust vectoring, and distributed reaction jets. Four compressor-nozzle outports located near the nose provide stabilizing moments during vertical liftoff, autonomous landing, and post-stall flight. In these low- q regimes the vehicle behaves more like a multirotor drone than a conventional fixed-wing aircraft, with reaction jets and thrust vectoring supplying the dominant control authority.

.4.2 D.2 Aerodynamic Surfaces

The lifting surfaces are based on F-22 style wings and stabilators, modified for SRBM:

- zero dihedral,

- symmetric planform,
- no static stability margin,
- no inherent lift vector,
- lift generated through commanded angle of attack,
- symmetric, speed-dependent bidirectional camber.

This ensures that all aerodynamic forces are intent-driven rather than geometry-driven, consistent with SRBM's rigid-body abstraction.

.4.3 D.3 Control Surfaces and Effectors

The airframe includes the following effectors:

D.3.1 Stabilators (Left and Right)

Provide pitch and roll authority and remain effective at high angle of attack due to symmetry.

D.3.2 Canted Rudders (Four Total)

Two dorsal and two ventral rudders provide:

- yaw authority,
- direct negative- X -axis drag braking,
- effectiveness in post-stall regimes.

D.3.3 Thrust-Vectoring Nozzle

A single engine with a thrust-vectoring nozzle:

- is aligned through the center of gravity,
- provides a 22° vectoring cone,
- enables pitch, yaw, and limited roll authority,
- supports VTOL, hover, and post-stall pivoting.

D.3.4 Forward Compressor Nozzles (Four Total)

Four compressor-driven reaction jets located near the nose provide:

- stabilizing pitch and yaw moments during vertical liftoff,
- attitude control during autonomous landing and descent,
- full-authority control in post-stall and zero-airspeed regimes,
- translational control during hover and precision touchdown,
- continuous controllability when aerodynamic surfaces are ineffective.

These nozzles allow the vehicle to operate as a drone-like, reaction-jet- stabilized rigid body whenever dynamic pressure is insufficient for aerodynamic control.

.4.4 D.4 Propulsion and Structural Capability

The propulsion system provides:

- thrust-to-weight ratio of approximately 1.5 : 1,
- sustained vertical climb and hover capability,
- full-attitude maneuvering.

The structure is rated for ± 20 G maneuvering, enabled by the absence of human physiological limits.

.4.5 D.5 Autonomous Flight Operations

The vehicle supports fully autonomous flight from liftoff to landing:

- **Autonomous vertical liftoff:** Compressor nozzles and thrust vectoring stabilize the vehicle through the zero-airspeed regime, with drone-like rigid-body control.
- **Automatic transition to forward flight:** Control authority shifts from reaction jets to aerodynamic surfaces as dynamic pressure increases.
- **AI-native flight:** All maneuvering is performed by the SRBM control stack; no pilot-centric modes or manual overrides are assumed.
- **Autonomous precision landing:** Reaction jets and thrust vectoring manage descent, attitude, and touchdown, again using drone-like control laws in the low- q regime.

.4.6 D.6 Actuator Vector

The actuator vector for this airframe is:

$$\mathbf{u}_{act} = (\delta_{stab,L}, \delta_{stab,R}, \delta_{rud,D1}, \delta_{rud,D2}, \delta_{rud,V1}, \delta_{rud,V2}, \delta_{TVC,x}, \delta_{TVC,y}) \quad (49)$$

(Forward compressor-nozzle commands exist physically but are omitted from this illustrative B_x structure for stability and clarity.)

.4.7 D.7 Control-Effectiveness Matrix Structure

Rigid-body moments are defined as:

$$\mathbf{r}_{mom} = (L, M, N) \quad (50)$$

The control-effectiveness matrix B_x for this example airframe has the original, stable symmetric structure:

$$B_x = \begin{bmatrix} +a & -a & 0 & 0 & 0 & 0 & b_x & 0 \\ +c & +c & 0 & 0 & 0 & 0 & 0 & b_y \\ 0 & 0 & d & d & -d & -d & 0 & 0 \end{bmatrix} \quad (51)$$

where a , c , d , b_x , and b_y are airframe-specific control-effectiveness coefficients identified from CFD, wind-tunnel testing, or flight data.

This matrix is illustrative only; SRBM does not require a specific structure. Its symmetry demonstrates the architectural requirement that tactical reasoning be fully decoupled from aerodynamic assumptions and privileged lift orientations.

.4.8 D.8 Integration with SRBM

This example demonstrates how an airframe instantiates SRBM:

- Layer 3 outputs normalized rigid-body commands (p, q, r, T, B) ,
- Layer 2 enforces feasibility and envelope limits,
- Layer 1 uses the airframe's B_x to compute actuator deflections.

SRBM remains portable across airframes with different geometries, provided they supply a valid B_x and actuator constraint set.